

G: Math United FC

Source file name: `mathunited.c`, `mathunited.cpp`, `mathunited.java`, or `mathunited.py`

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Math United FC is the football club with the most powerful data analytics department in the entire world. For that reason, it is the most wanted team for starting and experienced players worldwide.

You are the head of the player trading division which is in charge of hiring players. This division starts the player transfer season with a budget $B \geq 0$ that should never become negative due to financial fair play regulations. Using this budget, the team can hire two kinds of players:

1. **Commercial players:** players that in their current state bring revenue to the club. Hiring them increases the budget of the team. A commercial player has a strictly positive value.
2. **Football prospects:** players that have amazing talent, but currently will not generate any revenue for the club if hired. In fact, hiring them reduces or does not affect the overall budget of the club. Therefore, a prospect has a zero or negative value.

Your work in the trading division consists of looking at a long list of players of the two kinds to decide whether to hire or not a player with the available budget. Your hiring strategy should maximize the number of football prospects that are hired. The market is too competitive, so that, if you wait, then the player will be lost to another team. With these requirements in mind, you look at the whole list and tell each player if he/she is hired or not, in the same order he/she appears in the list.

For instance, suppose a budget of 9 and a list of 6 players, say A, B, C, D, E, F , whose values are $-4, -7, 5, -6, -1, -4$, respectively. Since you must decide on-the-fly if you hire each player in the list, and you want to maximize the hired prospects, in this case you should hire the players A, C, E, F , i.e., 3 prospects (A, E, F) and a commercial one (C). Note that, as required, the budget is never negative as long as you take these hiring decisions in the same order of the original list.

Input

The input consists of several test cases. Each test case starts with a line with two number $0 \leq B \leq 10^9$ and $1 \leq P \leq 200000$ separated by a single space. These values represent respectively the initial budget and the number of players you need to process in order. The next line contains P numbers separated by a single space representing how hiring the i -th player $-10^9 \leq P_i \leq 10^9$ affects the budget B .

The last line of input will be `0 0` and should not be processed.

The input must be read from standard input.

Output

For each test case, print a single line with one number x , stating the number of prospects that can be hired with the initial budget.

The output must be written to standard output.

Sample Input	Sample Output
9 6	3
-4 -7 5 -6 -1 -4	4
10 6	3
-4 -3 5 -2 -1 -6	
0 5	
-5 10 -2 -2 -2	
0 0	